

EXECUTIVE SUMMARY: EMERGENCY DEPARTMENT OPERATIONAL ANALYSIS

This report evaluates the operational performance of the Emergency Department based on the analysis of 9,216 patient interactions. The objective is to identify the root causes of the current 4.99 out of 10 satisfaction rating and the 35.3-minute average wait time.

Insight 1: Structural Misalignment of Staffing and Demand

The data identifies clear peak arrival periods on Mondays (1,377 patients) and Saturdays (1,322 patients). Daily surges occur specifically at 11:00, 13:00, 19:00, and 23:00.

- **Implications:** Current labor allocation is not synchronized with these predictable spikes in patient volume. This lack of alignment forces a limited number of staff to manage high-density arrival windows, which directly causes the 35.3-minute wait time and increases the risk of staff exhaustion.
- **Action & Targeted KPIs:** Redesign shift rosters to ensure maximum staffing overlaps during the four identified peak hours.
 - **KPI Target:** Reduce average wait time from **35.3 minutes to under 30 minutes** (a 15% efficiency gain).
 - **KPI Target:** Achieve a **0% variance** between peak arrival hours and maximum staff presence on Mondays and Saturdays.

Insight 2: Resource Bottlenecks in Specialist Transitions

While 5,400 patients require no specialist intervention, 2,835 patients must be directed to other departments. General Practice (1,840) and Orthopedics (995) account for 65 percent of all specialist transitions.

- **Implications:** The efficiency of the Emergency Department is constrained by the response times of these two specific departments. If General Practice or Orthopedic specialists are slow to accept a patient, emergency beds remains occupied, preventing new arrivals from being moved out of the waiting area.
- **Action & Targeted KPIs:** Establish a priority consultation protocol with the General Practice and Orthopedics departments.
 - **KPI Target:** Reduce the referral-to-transfer time for the **2,835 patients** requiring specialist care.
 - **KPI Target:** Aim for a **20% increase in bed turnover rate** by clearing "exit blocks" caused by these high-volume referral departments.

Insight 3: Service Gaps within the Primary Patient Demographic The largest patient cohort is aged between 20 and 39 years, totaling 2,388 individuals. This group is coupled with a highly diverse ethnic profile, including 1,951 African American and 1,557 Multiracial patients.

- **Implications:** A satisfaction score of 4.99 out of 10 indicates a significant failure to meet the expectations of our most frequent users. The current service model is failing

to provide the speed or communication clarity required by this young and diverse demographic, which may lead to a loss of community trust.

- **Action & Targeted KPIs:** Implement a digital communication platform that sends automated text updates to patients regarding their expected wait times.
 - **KPI Target:** Increase the Patient Satisfaction Score from **4.99 to a minimum of 7.0 out of 10** within the 20–39 age bracket.
 - **KPI Target:** Achieve a **90% positive feedback rate** on "Communication Clarity" in post-discharge surveys for the diverse patient base.

The Emergency Department is effectively managing a balanced ratio of 4,612 admissions and 4,604 releases. However, operational success is hindered by rigid staffing and specialist delays. Implementing these three targeted actions will reduce wait times and raise satisfaction scores to meet institutional targets.